

TITLE PAGE

Problem Statement: AI-Powered Behavioral Anomaly Detection for Cybersecurity

Theme: Cybersecurity / Industrial Security

PS Category: Software

Student Name: Atharv Agarwal

Student ID: 20522599

ARGUS

Adaptive Risk & Graph-based Unified Security

Precision 92.13% • Recall 100% • FP Rate 0.32%

Live: <https://arguskavachh.streamlit.app/>

ARGUS: Adaptive Risk & Graph-based Unified Security

PROPOSED SOLUTION

A UEBA-style system that learns per-entity & peer-group 'normal' behaviour from access logs and flags real-time deviations across 7 attack types — no signature rules. Fuses a sequence Transformer, a graph neural network, an Isolation Forest, deterministic hard rules, and a learned calibration layer.

HOW IT ADDRESSES THE PROBLEM

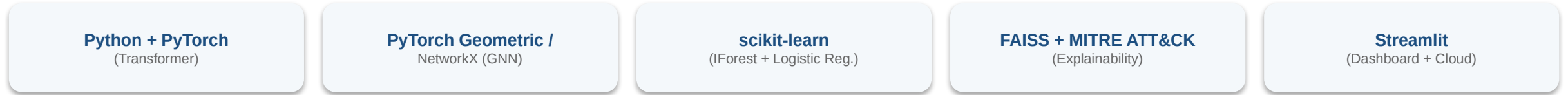
Sequential data → Transformer models session-order behaviour
Class imbalance → hard rules independent of skewed model confidence
Concept drift → dedicated monitor separates org change from attack
Explainability → alerts cite MITRE ATT&CK technique + behavioural deltas
Cold-start → peer-group baseline + Isolation Forest fallback

INNOVATION & UNIQUENESS

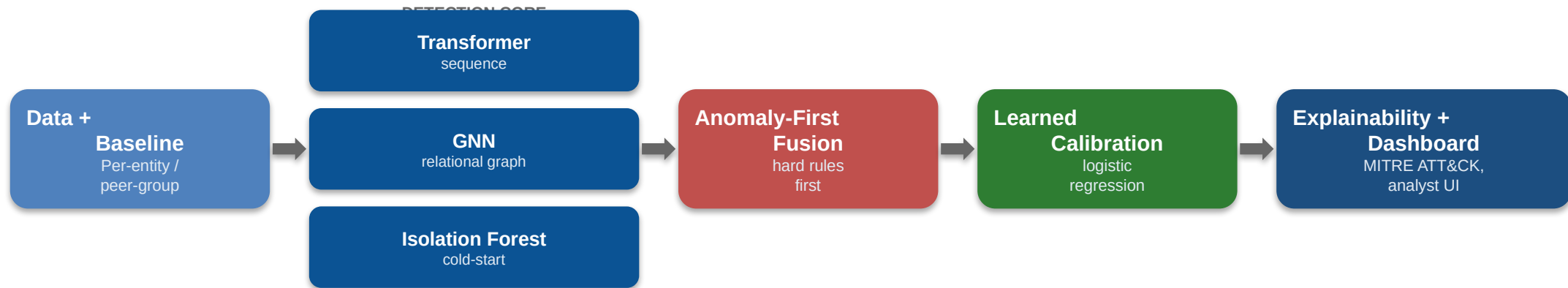
- Anomaly-first fusion: hard rules checked BEFORE model confidence
- 2 strong-looking models independently REJECTED after SHAP + calibration checks found shortcuts, not real signal
- Interpretable 3-coefficient calibration lifted precision 0.745 → 0.921 at unchanged 100% recall
- Architecture extensible to Honeywell Forge Cybersecurity+ / OT telemetry

TECHNICAL APPROACH

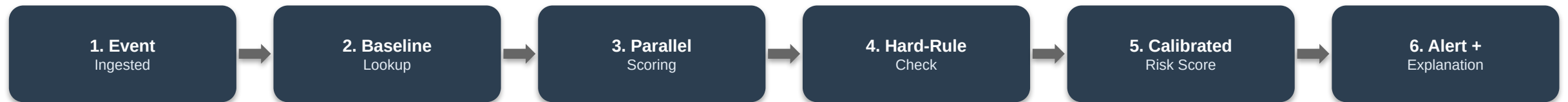
TECHNOLOGIES USED



SYSTEM ARCHITECTURE



PROCESS FLOW — single event, end to end



FEASIBILITY AND VIABILITY

FEASIBILITY

Fully working, deployed prototype — not just a concept (live link on Slide 6)

Verified on held-out test data: 92.13% precision, 100% recall, 0.32% FP rate across all 7 attack types

Built end-to-end (data → models → dashboard) within the hackathon window using a documented synthetic dataset

CHALLENGE: Rare attack classes have very few test samples

STRATEGY: Deterministic hard rules give robust detection independent of model confidence for these classes

CHALLENGE: Models can find shortcuts instead of real signals

STRATEGY: Verification discipline: SHAP attribution + calibration checks before adopting any result — 2 strong-looking models were caught and rejected

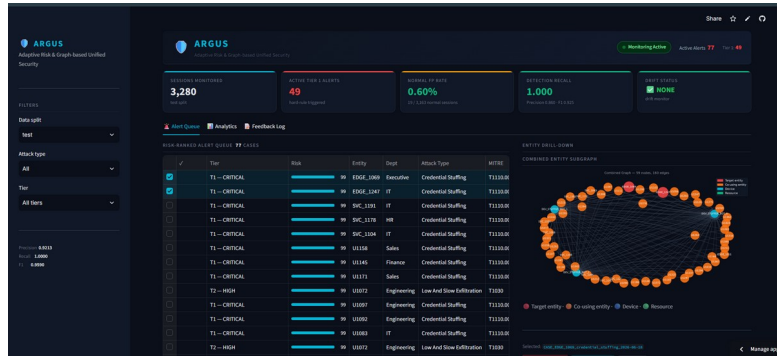
CHALLENGE: Scaling to real enterprise / OT data

STRATEGY: Modular fusion layer; architecture designed to extend to OT/ICS telemetry (Honeywell Forge alignment)

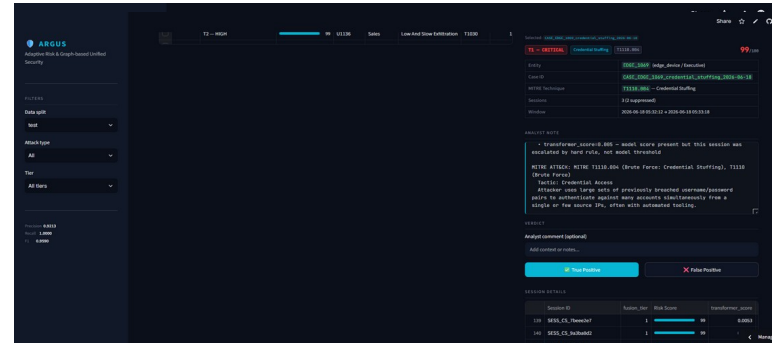
CHALLENGE: Legitimate behaviour drifts over time

STRATEGY: Dedicated drift monitor distinguishes 'organisation changed' from 'under attack'

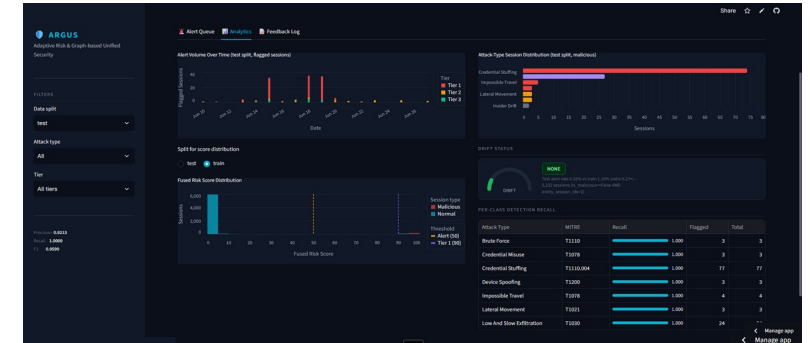
ARTIFACTS



Risk-ranked alert queue + entity subgraph



Explainability: MITRE ATT&CK-grounded analyst note



Analytics: score distribution & per-class recall

```
if hard_rule_fired(session):  
    # checked BEFORE model confidence  
    tier = 1  
    # geo-velocity / fp-mismatch / IP fan-in  
else:  
    score = logistic_calibration(transformer_score, if_norm, graph_boost)
```

Core logic — `anomaly_first_fusion.py`

Full source & solution proposal

GitHub: github.com/atharv1945/Argus

Live app: arguskavachh.streamlit.app

RESEARCH AND REFERENCES

Live Dashboard : <https://arguskavachh.streamlit.app/>

Source Code (GitHub) : github.com/atharv1945/Argus

MITRE ATT&CK Framework : attack.mitre.org — technique taxonomy grounding the explainability layer

Honeywell Forge Cybersecurity+ / Cyber Proactive Defense : Real-world OT/ICS behavioural-baselining products this architecture is designed to extend towards

UEBA industry background : Reviewed alongside commercial UEBA approaches (e.g. Exabeam, Splunk, Microsoft Sentinel) to ground the problem space